

constexpr std::shared_ptr

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Contents

1	Revision History	2
2	Introduction	2
3	Motivation and Scope	2
3.1	Atomic Operations	3
3.2	Two Memory Allocations	3
3.3	Relational Operators	4
3.4	Maybe Not Now, But Soon	5
4	Impact on the Standard	5
5	Implementation	5
6	Proposed Wording	5
7	Acknowledgements	11
8	References	12

1 Revision History

- R4 2024-10-21
 - Added `constexpr` qualifier to the wording for `std::bad_weak_ptr` with [P3068R4]
 - Updated ClangOz reference in motivation
- R3 2024-09-03
 - Removed `constexpr` specification from `reinterpret_pointer_cast` (see below [here](#))
 - Added references to [P3309R1] and [P3068R4]
 - Added details of a second implementation based on libc++
- R2 2024-05-24
 - Added wording
 - Removed `constexpr` specification from some functions (see below [here](#))
 - Removed SG7 from Audience (post 2024 Spring meeting in Tokyo)
- R1 2024-03-05
 - Added a motivating example
 - Included libc++ & MSVC STL in atomic operation considerations
- R0 2023-11-06
 - Original Proposal

2 Introduction

Since the adoption of [P0784R7] in C++20, constant expressions can include dynamic memory allocation; yet support for smart pointers extends only to `std::unique_ptr` (since [P2273R3] in C++23). As at runtime, smart pointers can encourage hygienic memory management during constant evaluation; and with no remaining technical obstacles, parity between runtime and compile-time support for smart pointers should reflect the increased maturity of language support for constant expression evaluation. We therefore propose that `std::shared_ptr` and appropriate class template member functions from 20.3 [smartptr] permit `constexpr` evaluation.

3 Motivation and Scope

It is convenient when the same C++ code can be deployed both at runtime and compile time. Our recent project investigates performance scaling of *parallel* constant expression evaluation in an experimental Clang compiler [ClangOz]. As well as C++17 parallel algorithms, a prototype `constexpr` implementation of the Khronos SYCL API was utilised, where a SYCL `buffer` class abstracts over device and/or host memory. In the simplified code excerpt below, the `std::shared_ptr` data member ensures memory is properly deallocated upon the `buffer`'s destruction, according to its owner status. This is a common approach for runtime code, and a `constexpr std::shared_ptr` class implementation helpfully bypasses thoughts of raw pointers and preprocessor macros. The impact of adding `constexpr` functionality to the SYCL implementation is therefore minimised.

```
template <class T, int dims = 1>
struct buffer
{
    constexpr buffer(const range<dims> &r)
        : range_{ r }, data_{ new T[r.size()], [this](auto* p){ delete [] p; } } { }

    constexpr buffer(T* hostData, const range<dims>& r)
        : range_{ r }, data_{ hostData, [] (auto){} } { }

    const range<dims> range_{ };
    std::shared_ptr<T[]> data_{ };
};
```

Adopted C++26 proposal [P2738R1] facilitates a straightforward implementation of comprehensive `constexpr` support for `std::shared_ptr`, allowing the `get_deleter` member function to operate, given the type erasure re-

quired within the `std::shared_ptr` unary class template. We furthermore propose that the relational operators of `std::unique_ptr`, which can legally operate on pointers originating from a single allocation during constant evaluation, should also adopt the `constexpr` specifier.

As with C++23 `constexpr` support for `std::unique_ptr`, bumping the value `__cpp_lib_constexpr_memory` is our requested feature macro change; yet in the discussion and implementation presented here, we adopt the macro `__cpp_lib_constexpr_shared_ptr`.

We below elaborate on points which go beyond the simple addition of the `constexpr` specifier to the relevant member functions.

3.1 Atomic Operations

The existing `std::shared_ptr` class can operate within a multithreaded runtime environment. A number of its member functions may therefore be defined using atomic functions; so ensuring that shared state is updated correctly. Atomic functions are not qualified as `constexpr`; but as constant expressions must be evaluated by a single thread, a `constexpr` `std::shared_ptr` implementation can safely skip calls to atomic functions through the predication of `std::is_constant_evaluated` (or `if constexpr`). For example, here is a modified function from GCC's libstdc++, called from `std::shared_ptr::use_count()` and elsewhere:

```
constexpr long
_M_get_use_count() const noexcept
{
#ifdef __cpp_lib_constexpr_shared_ptr
    return std::is_constant_evaluated()
        ? _M_use_count
        : __atomic_load_n(&_M_use_count, __ATOMIC_RELAXED);
#else
    return __atomic_load_n(&_M_use_count, __ATOMIC_RELAXED);
#endif
}
```

The use of atomic intrinsics within Clang's libc++ and MSVC's STL can be elided similarly. In `__memory/shared_ptr.h`, libc++ makes calls to the atomic intrinsic `__atomic_load_n`, only via the inline C++ functions `__libcxx_relaxed_load` and `__libcxx_acquire_load`; while `__atomic_add_fetch` is accessed only via `__libcxx_atomic_refcount_increment` and `__libcxx_atomic_refcount_decrement`. Each of these four functions is comprised only of return statement pairs, predicated upon *object-like* macros including `_LIBCPP_HAS_NO_THREADS`; and so could easily be modified to involve `std::is_constant_evaluated` as above.

In `stl/inc/memory`, the `std::shared_ptr` of MSVC's STL inherits a `_Ref_count_base` member through `_Ptr_base`. `_Ref_count_base` has two `_Atomic_counter_t` members (aliases of `unsigned long`), updated atomically using the `_InterlockedCompareExchange`, `_InterlockedIncrement` (via the macro `_MT_INCR`); or `_InterlockedDecrement` (via the macro `_MT_DECR`) atomic intrinsics. All the (five) functions invoking these intrinsics can again make use of `std::is_constant_evaluated` to avoid the atomic operations.

Adding `constexpr` support to an implementation of `std::shared_ptr` utilising `std::atomic` would need to take an alternative approach; likely involving the modification of its `std::atomic` definition. Recently, [P3309R1] has proposed adding such `constexpr` functionality to `std::atomic` (and `std::atomic_ref`) for C++26.

3.2 Two Memory Allocations

Unlike `std::unique_ptr`, a `std::shared_ptr` must store not only the managed object, but also the type-erased deleter and allocator, as well as the number of `std::shared_ptr`s and `std::weak_ptr`s which own or refer to the managed object. This information is managed as part of a dynamically allocated object referred to as the *control block*.

Existing runtime implementations of `std::make_shared`, `std::allocate_shared`, `std::make_shared_for_overwrite`, and `std::allocate_shared_for_overwrite`, allocate memory for both

the control block, *and* the managed object, from a single dynamic memory allocation; via `reinterpret_cast`. This practise aligns with a remark at 20.3.2.2.7 [\[util.smartptr.shared.create\]](#); quoted below:

- (7.1) — Implementations should perform no more than one memory allocation.
 — [Note 1: This provides efficiency equivalent to an intrusive smart pointer. — *end note*]

As `reinterpret_cast` is not permitted within a constant expression, an alternative approach is required for `std::make_shared`, `std::allocate_shared`, `std::make_shared_for_overwrite`, and `std::allocate_shared_for_overwrite`. A straightforward solution is to create the object first, and pass its address to the appropriate `std::shared_ptr` constructor. Considering the control block, this approach amounts to two dynamic memory allocations; albeit at compile-time. Assuming that the runtime implementation need not change, the remark quoted above can be left unchanged; as this is only a recommendation, not a requirement.

3.3 Relational Operators

Comparing dynamically allocated pointers within a constant expression is legal, provided the result of the comparison is not unspecified. Such comparisons are defined in terms of a partial order, applicable to pointers which either point “to different elements of the same array, or to subobjects thereof”; or to “different non-static data members of the same object, or to subobjects of such members, recursively...”; from paragraph 4 of 7.6.9 [\[expr.rel\]](#). A simple example program is shown below:

```
constexpr bool ptr_compare()
{
    int* p = new int[2]{};
    bool b = &p[0] < &p[1];
    delete [] p;
    return b;
}

static_assert(ptr_compare());
```

It is therefore unsurprising that we include the `std::shared_ptr` relational operators within the scope of our proposal to apply `constexpr` to all functions within 20.3 [\[smartptr\]](#); the `std::shared_ptr` aliasing constructor makes this especially simple to configure:

```
constexpr bool sptr_compare()
{
    double *arr = new double[2];
    std::shared_ptr p{&arr[0]}, q{p, p.get() + 1};
    return p < q;
}

static_assert(sptr_compare());
```

Furthermore, in the interests of `constexpr` consistency, we propose that the relational operators of `std::unique_ptr` *also* now include support for constant evaluation. As discussed above, the results of such comparisons are very often well defined.

It may be argued that a `std::unique_ptr` which is the sole owner of an array, or an object with data members, presents less need for relational operators. Yet we must consider that a custom deleter can easily change the operational semantics; as demonstrated in the example below. A `std::unique_ptr` should also be legally comparable with itself.

```
constexpr bool uptr_compare()
{
    short* p = new short[2]{};
    auto del = [](short*){};
    std::unique_ptr<short[]> a{p+0};
```

```

    std::unique_ptr<short[],decltype(del)> b{p+1, del};
    return a < b;
}

static_assert(uptr_compare());

```

3.4 Maybe Not Now, But Soon

The functions from 20.3 [smartptr] listed below cannot possibly be evaluated within a constant expression. We *do not* propose that their specifications should change. While C++23’s [P2448R2] allows such functions to be annotated as `constexpr`, we suggest that in this instance the C++ community will be served better by a future update; when their constant evaluation becomes possible.

- 20.3.3 [util.smartptr.hash]: The `operator()` member of the class template specialisations for `std::hash<std::unique_ptr<T,D>>` and `std::hash<std::shared_ptr<T>>` cannot be defined according to the *Cpp17Hash* requirements (16.4.4.5 [hash.requirements]). (A pointer cannot, during constant evaluation, be converted to an `std::size_t` using `reinterpret_cast`; or otherwise.)
- 20.3.2.5 [util.smartptr.owner.hash]: The two `operator()` member functions of the recently adopted `owner_hash` class, also cannot be defined according to the *Cpp17Hash* requirements.
- 20.3.2.2.6 [util.smartptr.shared.obs]: The recently adopted `owner_hash()` member function of `std::shared_ptr`, also cannot be defined according to the *Cpp17Hash* requirements.
- 20.3.2.3.6 [util.smartptr.weak.obs]: The recently adopted `owner_hash()` member function of `std::weak_ptr`, also cannot be defined according to the *Cpp17Hash* requirements.
- 20.3.2.2.10 [util.smartptr.shared.cast]: Neither of the two `reinterpret_pointer_cast` overloads can be included as their implementations will typically call `reinterpret_cast`, which is prohibited here.

We also *do not* propose any specification change for the overloads of `operator<<` for `std::shared_ptr` and `std::unique_ptr`, from 20.3.2.2.12 [util.smartptr.shared.io] and 20.3.1.7 [unique.ptr.io]. Unlike the functions above, a `constexpr` implementation for the overloads could today use a vendor-specific extension; do nothing; or simply report an error. But such possibilities should be discussed in a separate proposal focused on I/O.

4 Impact on the Standard

This proposal is a pure library extension, and does not require any new language features.

5 Implementation

An implementation based on the GNU C++ Library (libstdc++) can be found [here](#). A comprehensive test suite is included there within `tests/shared_ptr_constexpr_tests.cpp`; alongside a standalone bash script to run it. All tests pass with recent GCC and Clang (i.e. versions supporting [P2738R1]; `__cpp_constexpr >= 202306L`).

A second implementation, by Hana Dusíková, based on the “libc++” C++ Library is also available: on Github [here](#) (via commit 23217d0); and with a corresponding Compiler Explorer instance [here](#).

6 Proposed Wording

The following wording changes apply to [N4981] and can also be viewed on Github via a fork of the *C++ Standard Draft Sources* repository [here](#).

Add to 17.3.2 [version.syn] (Header `<version>` synopsis):

```

- #define __cpp_lib_constexpr_memory          202202L // freestanding, also in <memory>
+ #define __cpp_lib_constexpr_memory          YYYYMM // freestanding, also in <memory>

```

Add to 20.2.2 [memory.syn] (Header `<memory>` synopsis):

```

constexpr bool operator<(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator>(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator<=(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator>=(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr compare_three_way_result_t<typename unique_ptr<T1, D1>::pointer,
constexpr shared_ptr<T> make_shared(Args&&... args);
constexpr shared_ptr<T> allocate_shared(const A& a, Args&&... args);
constexpr shared_ptr<T> make_shared(size_t N);
constexpr shared_ptr<T> allocate_shared(const A& a, size_t N);
constexpr shared_ptr<T> make_shared();
constexpr shared_ptr<T> allocate_shared(const A& a);
constexpr shared_ptr<T> make_shared(size_t N, const remove_extent_t<T>& u);
constexpr shared_ptr<T> allocate_shared(const A& a, size_t N,
constexpr shared_ptr<T> make_shared(const remove_extent_t<T>& u);
constexpr shared_ptr<T> allocate_shared(const A& a, const remove_extent_t<T>& u);
constexpr shared_ptr<T> make_shared_for_overwrite();
constexpr shared_ptr<T> allocate_shared_for_overwrite(const A& a);
constexpr shared_ptr<T> make_shared_for_overwrite(size_t N);
constexpr shared_ptr<T> allocate_shared_for_overwrite(const A& a, size_t N);
constexpr bool operator==(const shared_ptr<T>& a, const shared_ptr<U>& b) noexcept;
constexpr strong_ordering operator<=(const shared_ptr<T>& a, const shared_ptr<U>& b) noexcept;
constexpr bool operator==(const shared_ptr<T>& x, nullptr_t) noexcept;
constexpr strong_ordering operator<=(const shared_ptr<T>& x, nullptr_t) noexcept;
constexpr void swap(shared_ptr<T>& a, shared_ptr<T>& b) noexcept;
constexpr shared_ptr<T> static_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> static_pointer_cast(shared_ptr<U>&& r) noexcept;
constexpr shared_ptr<T> dynamic_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> dynamic_pointer_cast(shared_ptr<U>&& r) noexcept;
constexpr shared_ptr<T> const_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> const_pointer_cast(shared_ptr<U>&& r) noexcept;
constexpr D* get_deleter(const shared_ptr<T>& p) noexcept;
template<class T> constexpr void swap(weak_ptr<T>& a, weak_ptr<T>& b) noexcept;
constexpr auto out_ptr(Smart& s, Args&&... args);
constexpr auto inout_ptr(Smart& s, Args&&... args);

```

Add to 20.3.1.6 [\[unique.ptr.special\]](#) (Specialized algorithms):

```

constexpr bool operator<(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator>(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator<=(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr bool operator>=(const unique_ptr<T1, D1>& x, const unique_ptr<T2, D2>& y);
constexpr compare_three_way_result_t<typename unique_ptr<T1, D1>::pointer,

```

Add to 20.3.2.1 [\[util.smartptr.weak.bad\]](#) (Class bad_weak_ptr):

```

constexpr const char* what() const noexcept override;
constexpr const char* what() const noexcept override;

```

Add to 20.3.2.2.1 [\[util.smartptr.shared.general\]](#) (General):

```

constexpr explicit shared_ptr(Y* p);
constexpr shared_ptr(Y* p, D d);
constexpr shared_ptr(Y* p, D d, A a);
constexpr shared_ptr(nullptr_t p, D d);
constexpr shared_ptr(nullptr_t p, D d, A a);
template<class Y> constexpr shared_ptr(const shared_ptr<Y>& r, element_type* p) noexcept;

```

```

template<class Y> constexpr shared_ptr(shared_ptr<Y>&& r, element_type* p) noexcept;
constexpr shared_ptr(const shared_ptr& r) noexcept;
template<class Y> constexpr shared_ptr(const shared_ptr<Y>& r) noexcept;
constexpr shared_ptr(shared_ptr&& r) noexcept;
template<class Y> constexpr shared_ptr(shared_ptr<Y>&& r) noexcept;
template<class Y> constexpr explicit shared_ptr(const weak_ptr<Y>& r);
constexpr shared_ptr(unique_ptr<Y, D>&& r);
constexpr ~shared_ptr();
constexpr shared_ptr& operator=(const shared_ptr& r) noexcept;
template<class Y> constexpr shared_ptr& operator=(const shared_ptr<Y>& r) noexcept;
constexpr shared_ptr& operator=(shared_ptr&& r) noexcept;
template<class Y> constexpr shared_ptr& operator=(shared_ptr<Y>&& r) noexcept;
constexpr shared_ptr& operator=(unique_ptr<Y, D>&& r);
constexpr void swap(shared_ptr& r) noexcept;
constexpr void reset() noexcept;
constexpr void reset(Y* p);
constexpr void reset(Y* p, D d);
constexpr void reset(Y* p, D d, A a);
constexpr element_type* get() const noexcept;
constexpr T& operator*() const noexcept;
constexpr T* operator->() const noexcept;
constexpr element_type& operator[](ptrdiff_t i) const;
constexpr long use_count() const noexcept;
constexpr explicit operator bool() const noexcept;
template<class U> constexpr bool owner_before(const shared_ptr<U>& b) const noexcept;
template<class U> constexpr bool owner_before(const weak_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const shared_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const weak_ptr<U>& b) const noexcept;

```

Add to 20.3.2.2.2 [\[util.smartptr.shared.const\]](#) (Constructors):

```

template<class Y> constexpr explicit shared_ptr(Y* p);
template<class Y, class D> constexpr shared_ptr(Y* p, D d);
template<class Y, class D, class A> constexpr shared_ptr(Y* p, D d, A a);
template<class D> constexpr shared_ptr(nullptr_t p, D d);
template<class D, class A> constexpr shared_ptr(nullptr_t p, D d, A a);
template<class Y> constexpr shared_ptr(const shared_ptr<Y>& r, element_type* p) noexcept;
template<class Y> constexpr shared_ptr(shared_ptr<Y>&& r, element_type* p) noexcept;
constexpr shared_ptr(const shared_ptr& r) noexcept;
template<class Y> constexpr shared_ptr(const shared_ptr<Y>& r) noexcept;
constexpr shared_ptr(shared_ptr&& r) noexcept;
template<class Y> constexpr shared_ptr(shared_ptr<Y>&& r) noexcept;
template<class Y> constexpr explicit shared_ptr(const weak_ptr<Y>& r);
template<class Y, class D> constexpr shared_ptr(unique_ptr<Y, D>&& r);

```

Add to 20.3.2.2.3 [\[util.smartptr.shared.dest\]](#) (Destructor):

```

constexpr ~shared_ptr();

```

Add to 20.3.2.2.4 [\[util.smartptr.shared.assign\]](#) (Assignment):

```

constexpr shared_ptr& operator=(const shared_ptr& r) noexcept;
template<class Y> constexpr shared_ptr& operator=(const shared_ptr<Y>& r) noexcept;
constexpr shared_ptr& operator=(shared_ptr&& r) noexcept;
template<class Y> constexpr shared_ptr& operator=(shared_ptr<Y>&& r) noexcept;
template<class Y, class D> constexpr shared_ptr& operator=(unique_ptr<Y, D>&& r);

```


Add to 20.3.2.2.5 [\[util.smartptr.shared.mod\]](#) (Modifiers):

```
constexpr void swap(shared_ptr& r) noexcept;  
constexpr void reset() noexcept;  
template<class Y> constexpr void reset(Y* p);  
template<class Y, class D> constexpr void reset(Y* p, D d);  
template<class Y, class D, class A> constexpr void reset(Y* p, D d, A a);
```

Add to 20.3.2.2.6 [\[util.smartptr.shared.obs\]](#) (Observers):

```
constexpr element_type* get() const noexcept;  
constexpr T& operator*() const noexcept;  
constexpr T* operator->() const noexcept;  
constexpr element_type& operator[](ptrdiff_t i) const;  
constexpr long use_count() const noexcept;  
constexpr explicit operator bool() const noexcept;  
template<class U> constexpr bool owner_before(const shared_ptr<U>& b) const noexcept;  
template<class U> constexpr bool owner_before(const weak_ptr<U>& b) const noexcept;  
constexpr bool owner_equal(const shared_ptr<U>& b) const noexcept;  
constexpr bool owner_equal(const weak_ptr<U>& b) const noexcept;
```

Add to 20.3.2.2.7 [\[util.smartptr.shared.create\]](#) (Creation):

```
constexpr shared_ptr<T> make_shared(argc);  
constexpr shared_ptr<T> allocate_shared(const A& a, argc);  
constexpr shared_ptr<T> make_shared_for_overwrite(argc);  
constexpr shared_ptr<T> allocate_shared_for_overwrite(const A& a, argc);  
constexpr shared_ptr<T> make_shared(Args&&... args);  
constexpr shared_ptr<T> allocate_shared(const A& a, Args&&... args);  
constexpr make_shared(size_t N);  
constexpr shared_ptr<T> allocate_shared(const A& a, size_t N);  
constexpr shared_ptr<T> make_shared();  
constexpr shared_ptr<T> allocate_shared(const A& a);  
constexpr shared_ptr<T> make_shared(size_t N, const remove_extent_t<T>& u);  
constexpr shared_ptr<T> allocate_shared(const A& a, size_t N, const remove_extent_t<T>& u);  
constexpr shared_ptr<T> make_shared(const remove_extent_t<T>& u);  
constexpr shared_ptr<T> allocate_shared(const A& a, const remove_extent_t<T>& u);  
constexpr shared_ptr<T> make_shared_for_overwrite();  
constexpr shared_ptr<T> allocate_shared_for_overwrite(const A& a);  
constexpr shared_ptr<T> make_shared_for_overwrite(size_t N);  
constexpr shared_ptr<T> allocate_shared_for_overwrite(const A& a, size_t N);
```

Add to 20.3.2.2.8 [\[util.smartptr.shared.cmp\]](#) (Comparison):

```
constexpr bool operator==(const shared_ptr<T>& a, const shared_ptr<U>& b) noexcept;  
template<class T> constexpr bool operator==(const shared_ptr<T>& a, nullptr_t) noexcept;  
constexpr strong_ordering operator<==(const shared_ptr<T>& a, const shared_ptr<U>& b) noexcept;  
constexpr strong_ordering operator<==(const shared_ptr<T>& a, nullptr_t) noexcept;
```

Add to 20.3.2.2.9 [\[util.smartptr.shared.spec\]](#) (Specialized algorithms):

```
constexpr void swap(shared_ptr<T>& a, shared_ptr<T>& b) noexcept;
```

Add to 20.3.2.2.10 [\[util.smartptr.shared.cast\]](#) (Casts):


```
constexpr shared_ptr<T> static_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> static_pointer_cast(shared_ptr<U>&& r) noexcept;
constexpr shared_ptr<T> dynamic_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> dynamic_pointer_cast(shared_ptr<U>&& r) noexcept;
constexpr shared_ptr<T> const_pointer_cast(const shared_ptr<U>& r) noexcept;
constexpr shared_ptr<T> const_pointer_cast(shared_ptr<U>&& r) noexcept;
```

Add to 20.3.2.2.11 [\[util.smartptr.getdeleter\]](#) (get_deleter):

```
constexpr D* get_deleter(const shared_ptr<T>& p) noexcept;
```

Add to 20.3.2.3.1 [\[util.smartptr.weak.general\]](#) (General):

```
constexpr weak_ptr(const shared_ptr<Y>& r) noexcept;
constexpr weak_ptr(const weak_ptr& r) noexcept;
constexpr weak_ptr(const weak_ptr<Y>& r) noexcept;
constexpr weak_ptr(weak_ptr&& r) noexcept;
constexpr weak_ptr(weak_ptr<Y>&& r) noexcept;
constexpr ~weak_ptr();
constexpr weak_ptr& operator=(const weak_ptr& r) noexcept;
constexpr weak_ptr& operator=(const weak_ptr<Y>& r) noexcept;
constexpr weak_ptr& operator=(const shared_ptr<Y>& r) noexcept;
constexpr weak_ptr& operator=(weak_ptr&& r) noexcept;
constexpr weak_ptr& operator=(weak_ptr<Y>&& r) noexcept;
constexpr void swap(weak_ptr& r) noexcept;
constexpr void reset() noexcept;
constexpr long use_count() const noexcept;
constexpr bool expired() const noexcept;
constexpr shared_ptr<T> lock() const noexcept;
constexpr bool owner_before(const shared_ptr<U>& b) const noexcept;
constexpr bool owner_before(const weak_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const shared_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const weak_ptr<U>& b) const noexcept;
```

Add to 20.3.2.3.2 [\[util.smartptr.weak.const\]](#) (Constructors):

```
constexpr weak_ptr(const weak_ptr& r) noexcept;
template<class Y> constexpr weak_ptr(const weak_ptr<Y>& r) noexcept;
template<class Y> constexpr weak_ptr(const shared_ptr<Y>& r) noexcept;
constexpr weak_ptr(weak_ptr&& r) noexcept;
template<class Y> constexpr weak_ptr(weak_ptr<Y>&& r) noexcept;
```

Add to 20.3.2.3.3 [\[util.smartptr.weak.dest\]](#) (Destructor):

```
constexpr ~weak_ptr();
```

Add to 20.3.2.3.4 [\[util.smartptr.weak.assign\]](#) (Assignment):

```
constexpr weak_ptr& operator=(const weak_ptr& r) noexcept;
template<class Y> constexpr weak_ptr& operator=(const weak_ptr<Y>& r) noexcept;
template<class Y> constexpr weak_ptr& operator=(const shared_ptr<Y>& r) noexcept;
constexpr weak_ptr& operator=(weak_ptr&& r) noexcept;
template<class Y> constexpr weak_ptr& operator=(weak_ptr<Y>&& r) noexcept;
```

Add to 20.3.2.3.5 [\[util.smartptr.weak.mod\]](#) (Modifiers):

```
constexpr void swap(weak_ptr& r) noexcept;
constexpr void reset() noexcept;
```

Add to 20.3.2.3.6 [\[util.smartptr.weak.obs\]](#) (Observers):

```
constexpr long use_count() const noexcept;
constexpr bool expired() const noexcept;
constexpr shared_ptr<T> lock() const noexcept;
template<class U> constexpr bool owner_before(const shared_ptr<U>& b) const noexcept;
template<class U> constexpr bool owner_before(const weak_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const shared_ptr<U>& b) const noexcept;
constexpr bool owner_equal(const weak_ptr<U>& b) const noexcept;
```

Add to 20.3.2.3.7 [\[util.smartptr.weak.spec\]](#) (Specialized algorithms):

```
constexpr void swap(weak_ptr<T>& a, weak_ptr<T>& b) noexcept;
```

Add to 20.3.2.4 [\[util.smartptr.ownerless\]](#) (Class template owner_less):

```
constexpr bool operator()(const shared_ptr<T>&, const shared_ptr<T>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>&, const weak_ptr<T>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const shared_ptr<T>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const weak_ptr<T>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>&, const weak_ptr<T>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const shared_ptr<T>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>&, const shared_ptr<U>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>&, const weak_ptr<U>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const shared_ptr<U>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const weak_ptr<U>&) const noexcept;
```

Add to 20.3.2.6 [\[util.smartptr.owner.equal\]](#) (Struct owner_equal):

```
constexpr bool operator()(const shared_ptr<T>&, const shared_ptr<U>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>&, const weak_ptr<U>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const shared_ptr<U>&) const noexcept;
constexpr bool operator()(const weak_ptr<T>&, const weak_ptr<U>&) const noexcept;
constexpr bool operator()(const shared_ptr<T>& x, const shared_ptr<U>& y) const noexcept;
constexpr bool operator()(const shared_ptr<T>& x, const weak_ptr<U>& y) const noexcept;
constexpr bool operator()(const weak_ptr<T>& x, const shared_ptr<U>& y) const noexcept;
constexpr bool operator()(const weak_ptr<T>& x, const weak_ptr<U>& y) const noexcept;
```

Add to 20.3.2.7 [\[util.smartptr.enab\]](#) (Class template enable_shared_from_this):

```
constexpr enable_shared_from_this(const enable_shared_from_this&) noexcept;
constexpr enable_shared_from_this& operator=(const enable_shared_from_this&) noexcept;
constexpr ~enable_shared_from_this();
constexpr shared_ptr<T> shared_from_this();
constexpr shared_ptr<T const> shared_from_this() const;
constexpr weak_ptr<T> weak_from_this() noexcept;
constexpr weak_ptr<T const> weak_from_this() const noexcept;
constexpr enable_shared_from_this(const enable_shared_from_this<T>&) noexcept;
constexpr enable_shared_from_this<T>& operator=(const enable_shared_from_this<T>&) noexcept;
constexpr shared_ptr<T> shared_from_this();
constexpr shared_ptr<T const> shared_from_this() const;
constexpr weak_ptr<T> weak_from_this() noexcept;
constexpr weak_ptr<T const> weak_from_this() const noexcept;
```

Add to 20.3.4.1 [\[out.ptr.t\]](#) (Class template out_ptr_t):

```
constexpr explicit out_ptr_t(Smart&, Args...);
constexpr ~out_ptr_t();
constexpr operator Pointer*() const noexcept;
constexpr operator void**() const noexcept;
constexpr explicit out_ptr_t(Smart& smart, Args... args);
constexpr ~out_ptr_t();
constexpr operator Pointer*() const noexcept;
constexpr operator void**() const noexcept;
```

Add to 20.3.4.2 [\[out.ptr\]](#) (Function template `out_ptr`):

```
constexpr auto out_ptr(Smart& s, Args&&... args);
```

Add to 20.3.4.3 [\[inout.ptr.t\]](#) (Class template `inout_ptr_t`):

```
constexpr explicit inout_ptr_t(Smart&, Args...);
constexpr ~inout_ptr_t();
constexpr operator Pointer*() const noexcept;
constexpr operator void**() const noexcept;
constexpr explicit inout_ptr_t(Smart& smart, Args... args);
constexpr ~inout_ptr_t();
constexpr operator Pointer*() const noexcept;
constexpr operator void**() const noexcept;
```

Add to 20.3.4.4 [\[inout.ptr\]](#) (Function template `inout_ptr`):

```
constexpr auto inout_ptr(Smart& s, Args&&... args);
```

7 Acknowledgements

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8 References

- [ClangOz] Andrew Gozillon. 2024. ClangOz: Parallel constant evaluation of C++ map and reduce operations.
<https://doi.org/10.1016/j.cola.2024.101298>
- [N4981] Thomas Köppe. 2024-04-16. Working Draft, Programming Languages — C++.
<https://wg21.link/n4981>
- [P0784R7] Daveed Vandevoorde. 2019. More constexpr containers.
<https://wg21.link/p0784>
- [P2273R3] Andreas Fertig. 2021. Making `std::unique_ptr` constexpr.
<https://wg21.link/p2273>
- [P2448R2] Barry Revzin. 2022. Relaxing some `constexpr` restrictions.
<https://wg21.link/p2448>
- [P2738R1] David Ledger. 2023. `constexpr` cast from `void*`: towards `constexpr` type-erasure.
<https://wg21.link/p2738>
- [P3068R4] Hana Dusíková. 2024. Allowing exception throwing in constant-evaluation.
<https://wg21.link/p3068>
- [P3309R1] Hana Dusíková. 2024. constexpr atomic and atomic_ref.
<https://wg21.link/p3309>